Model Report

1. Preprocessing Decisions and Text Cleaning

The SEC 10-K filings dataset was loaded from Hugging Face and processed to create a structured text classification pipeline. Since the **Risk Factors** section was largely empty in the sampled filings, the **Business** section and **Management's Discussion and Analysis of Financial Condition and Results of Operations (MD&A)** section were combined to form the primary text corpus.

The following preprocessing steps were applied:

- Extracted relevant textual sections from each filing.
- Combined Business and MD&A sections into a single document.
- Removed missing and null values.
- Converted text into a standardized format suitable for feature extraction.
- Generated document-length statistics for label creation.
- Limited the working dataset to 5,000 filings to ensure efficient model training and evaluation.

These steps ensured that the dataset contained meaningful textual information while reducing computational complexity.

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2. Features Used and Rationale

TF-IDF Features

Term Frequency-Inverse Document Frequency (TF-IDF) was used to convert text into numerical features.

Parameters used:

- `max_features = 5000`
- English stop-word removal enabled

Why TF-IDF?

TF-IDF was selected because:

- It efficiently represents textual information as numerical vectors.
- Frequently occurring but less informative words receive lower importance.
- Important domain-specific terms receive higher weights.
- It is computationally efficient and performs well for traditional machine learning models.
- It provides strong baseline performance for financial document classification tasks.

The resulting feature matrix was used as input to all classification models.

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3. Model Comparison and Evaluation

Three machine learning models were trained and evaluated:

1. AdaBoost Classifier
2. XGBoost Classifier
3. CatBoost Classifier

Performance Metrics

Model	Accuracy	Precision	Recall	F1 Score
AdaBoost	0.922	0.9226	0.922	0.9222
XGBoost	0.978	0.9780	0.978	0.9780
CatBoost	0.978	0.9781	0.978	0.9780

Interpretation

- AdaBoost achieved strong performance with an F1 score above 92%.
- XGBoost significantly improved classification accuracy and produced highly balanced precision and recall values.
- CatBoost achieved the highest overall performance with the best precision and F1 score among all models.

The high scores indicate that TF-IDF features effectively captured differences between document classes and that ensemble boosting methods were highly suitable for this classification task.

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4. Best Model Selection

Selected Model: CatBoost

CatBoost was selected as the final model because it achieved the highest overall performance across evaluation metrics.

Reasons for selection:

- Highest Precision Score (0.9781)
- Highest F1 Score (0.9780)
- Excellent balance between precision and recall
- Robust handling of high-dimensional text features
- Strong generalization performance on unseen data

Although XGBoost produced nearly identical results, CatBoost demonstrated slightly superior performance and was therefore chosen as the final deployment model.

The selected CatBoost model was serialized and deployed through a FastAPI-based REST API, enabling real-time document classification through the `/predict` endpoint.